THE MALAWI NATIONAL EXAMINATIONS BOARD

M131/II

2004 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

MATHEMATICS

Subject Number: M131/II

Time Allowed: 2 h 30 mins
8.30 - 11.00 am

Tuesday, 19 October

PAPER II
(100 marks)

Instructions:

1. **This paper contains 6 pages. Please check.**

2. Answer **all** the **six** questions in Section **A** and any **three** questions from Section **B**.

3. The maximum number of marks for each answer is indicated against each question.

4. Mathematical tables and answer books are provided.

5. Calculators may be used.

6. Used graph paper and/or supplementary sheets must be tied together inside the answer book with a string.

7. **All working must be clearly shown**; it should be done on the same sheet as the rest of the answers.

8. Write your **Examination Number** on top of each page of your Answer Book.

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Section A (55 marks)

Answer **all** the **six** questions in this section.

$\frac{-3a}{-8a-a} = \frac{a}{3a} = \frac{1}{3}$

1. a. If $f(x) = \frac{3x}{8x-a}$, find $f(-a)$ in its simplest form. **(4 marks)**

   b. The distance between points M(3,5) and N(8,a) is 13. Find the value of a. **(6 marks)**

   $\sqrt{5^2+(5-a)^2} = 13$ $\quad 50 - 2a + a^2 = 169$ $\quad a^2 - 2a - 119 = 0$ $\quad a = 1 \pm \sqrt{1+119}$ $\quad = 1 \pm 2\sqrt{30}$

2. The sum of the squares of two consecutive odd numbers is 74. If n is the smaller of the two numbers, find the values of n. **(7 marks)**

   $n + a(n+2) = 74$ $\quad \therefore n = 36$ //

   $2n = 72$

3. a. The sum of n terms of an arithmetic progression is $\frac{5n^2+3n}{2}$. $S_1 = a_1 = \frac{5+3}{2} = 4$

   Calculate the first two terms of the arithmetic progression. **(5 marks)**

   $a_2 = S_2 - S_1$

   $= \frac{20+6}{2} - 4 = 9$

   b. The position vector of points A, B, C and D are $\binom{0}{0}$, $\binom{4}{8}$, $\binom{6}{8}$ and $\binom{2}{-5}$ respectively.

   Show by vector method that ABCD is a parallelogram. **(5 marks)**

   not

   $\vec{AB} = \binom{4}{8}$

   $\vec{CD} =$

4. a. **Figure 1** show a graph of $y = ax^2 + bx + c$ where a, b and c are constants.

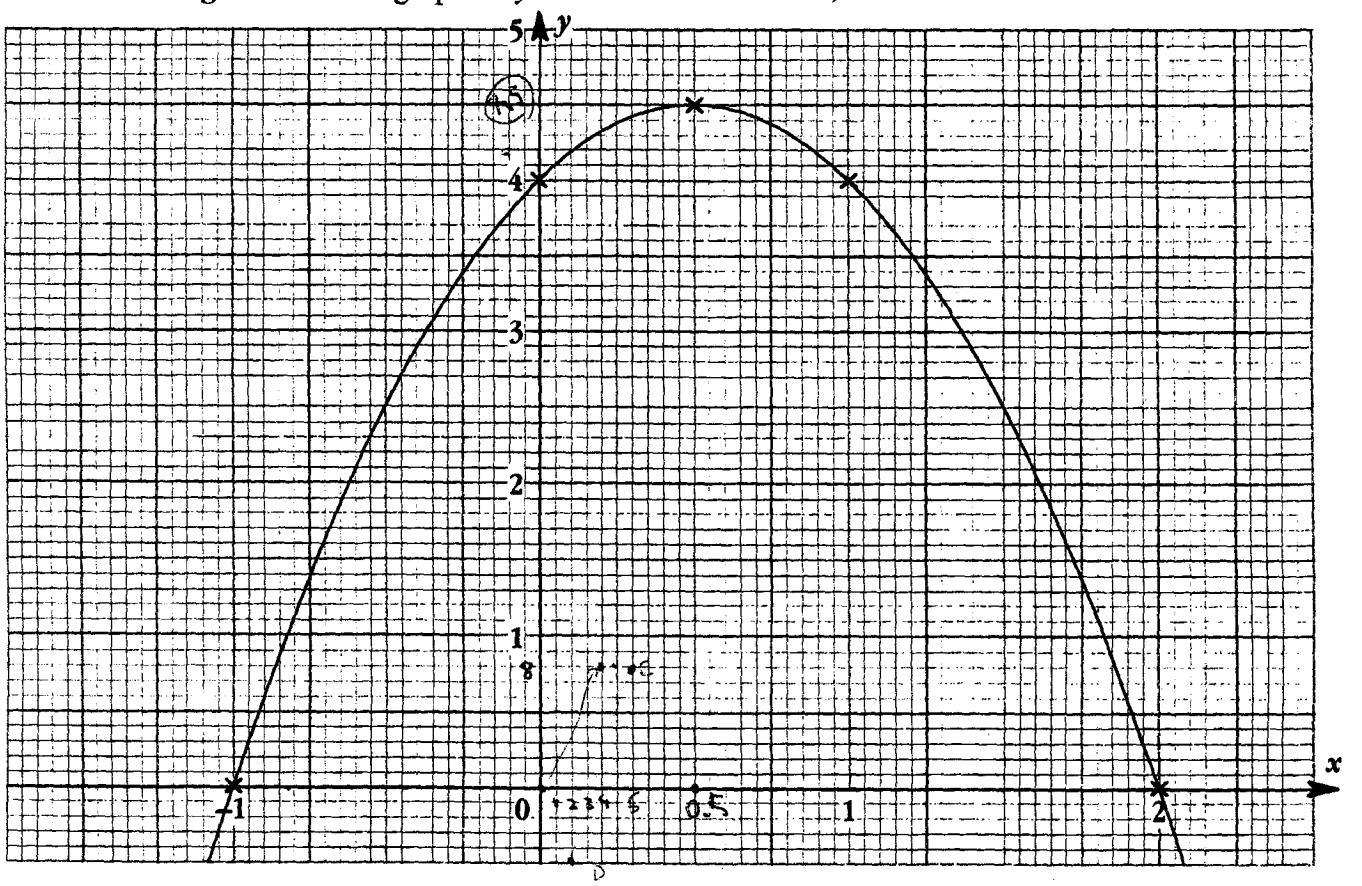

Figure 1

Find the equation of:

(i) the curve. **(5 marks)**

$(0,4)$, $(1,4)$, $(2,0)$ $\begin{cases} 4 = c \\ 4 = a+b+4 \\ 0 = 4a+2b+4 \end{cases}$

(ii) the line of symmetry. **(1 mark)**

$\Leftrightarrow \begin{cases} a+b = 0 \\ 2a+b = -2 \end{cases}$

$\Leftrightarrow \begin{cases} a = 2 \\ b = -2 \end{cases}$

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4. **(Continued)**

b. In **Figure 2,** triangle **LMN** is such that angle **LMN** = 60°, angle **MNL** = 45°, and **LM** = 2 cm.

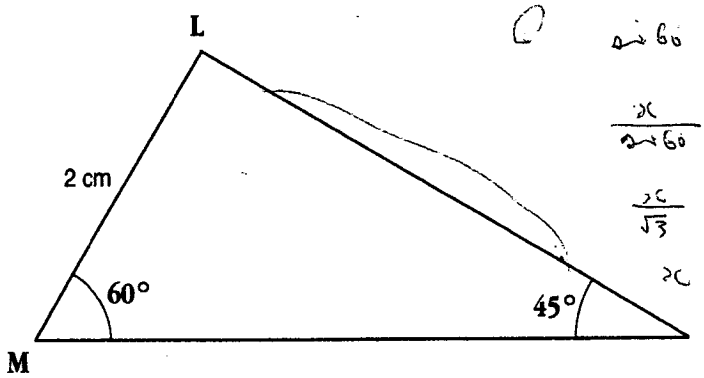


Figure 2

Calculate the length of **LN**, leaving your answer in surd form. **(5 marks)**

5. A die and a coin are rolled and tossed respectively.

a. Draw a probability space table. **(3 marks)**

b. Use the probability space table to find the probability of obtaining an even number on the die and a tail on the coin. **(2 marks)**

6. a. **Figure 3** shows a chord **AB** subtending an angle of 35° at **P** on the circumference of a circle centre **O.**

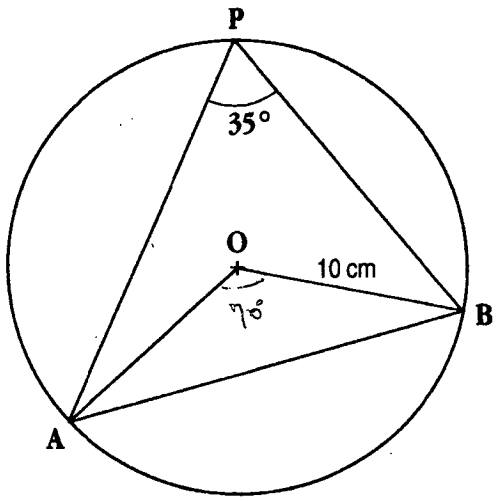


Figure 3

If the radius of the circle is 10 cm, calculate the length of the chord **AB**. **(5 marks)**

6. **(Continued)**

b. Using a ruler and a pair of compasses only, construct on the same diagram:

(i) a circle centre **O** radius 4 cm.

(ii) a tangent **PQ** at P such that **PQ** = 6 cm.

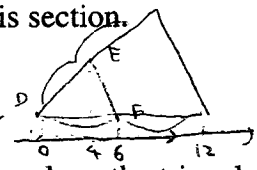
(iii) a line **OR** 10 cm long such that angle **POR** = 120° and PR is longer than **QR**.

(iv) a point **S** on the circumference of a circle closer to **Q** such that **RS** is a tangent to the circle.

(v) measure and state length of **QS**. **(7 marks)**

Section B (45 marks)

Answer any **three** questions from this section.



7. a. A triangle DEF has vertices D(0,2), E(4,6) and F(6,2).

(i) Using a scale of 2 cm to represent 2 units in both axes, draw the triangle DEF on the graph paper provided. **(2 marks)**

(ii) On the same axes draw the enlargement D′ E′ F′ using D(0,2) as centre of enlargement and a scale factor of 2. **(3 marks)**

b. A quantity q is the difference between two parts. The first part is constant and the second part varies inversely as the square of p. If $q = 1$ then $p = 2$ and $q = 6$ when $p = 3$. Find the positive value of p when $q = 9$. **(10 marks)**

$q = ap + b$ $\begin{cases} 1 = 2a + b \\ 6 = 3a + b \end{cases}$ $\begin{cases} a = 5 \\ b = -9 \end{cases}$ $q = 5p - 9$

$p = \frac{18}{5}$ //

8. a. Express $1 - \frac{1}{x} - \frac{x-2}{x-1}$ as a single fraction in its simplest form. **(4 marks)**

$= \frac{x(x-1) - (x-1) - x(x-2)}{x(x-1)} = \frac{\cancel{x^2} - \cancel{x} - \cancel{x} + 1 - \cancel{x^2} + 2x}{x(x-1)} = \frac{1}{x(x-1)}$

b. (i) Copy and complete the table of values for the equation $y = x(x^2 - x - 6)$.

x	-3	-2	-1	0	1	2	3	4
y	-18	0	4	0	-6	-8	0	24

$-(1+1-6) = 4$ $2(4-2-6) = -8$

(ii) Using a scale of 2 cm to represent 1 unit on the x-axis and 2 cm to represent 5 units on the y-axis, draw the graph of $y = x(x^2 - x - 6)$.

(iii) Use your graph to solve the equation $x(x^2 - x - 6) + 5 = 0$. **(11 marks)**

$2.5\,(2.5^2 - 2.5 - 6)$

$= 3.5\,(12.25 - 9.5)$

$= 3.5 \cdot 2.75$

$\begin{array}{r} 3.5 \\ 3.5 \\ \hline 17.5 \\ 105 \\ \hline 12.25 \end{array}$

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9. a. Given that $X = \{11, 12, 15, 17\}$
$Y = \{11, 15, 17, 19, 20\}$
$Z = \{11, 17, 21\}$

(i) Show the three sets on a venn diagram.

(ii) Find $n(X \cap Y \cap Z)$. 11, 17

(6 marks)

b. A boat sails 40 km from point A to point B, east of A on a lake. Lodge P is south of point A. Another lodge Q is south of point B. The bearing of Q from point A is S52°E and the bearing of P from B is S39°W.

If P is 60 km from Q, calculate the bearing of lodge Q from lodge P. **(9 marks)**

10. a. If $(x + 1)$ is a factor of $x^3 - 5x^2 + 2x + 8$,

(i) find the other **two** factors, hence

(ii) find the roots of the equation $x^3 - 5x^2 + 2x + 8 = 0$. **(7 marks)**

b. Calculate the standard deviation of the following numbers:

$x - 1, x, x + 1.$ **(8 marks)**

11. a. Solve the equation $\log_9 27^K = K + 1$. **(5 marks)**

b. Suppose you have K1 500 to spend in a shop. You have decided to:

- divide the money between buying bottles of cooking oil at K150 each and tablets of soap at K60 each.
- buy at least 5 tablets of soap.
- spend more on cooking oil than soap by more than K300.

(i) If x and y represent the number of bottles of cooking oil and tablets of soap respectively, write down **three** inequalities which x and y must satisfy the above three conditions.

(ii) Using a scale of 2 cm to represent 2 units on the x-axis and 2 cm to represent 5 units on the y-axis, draw a region bounded by the four inequalities obtained in (i) above by shading unwanted area.

(iii) For which values of x and y is $x + y$ the greatest? **(10 marks)**

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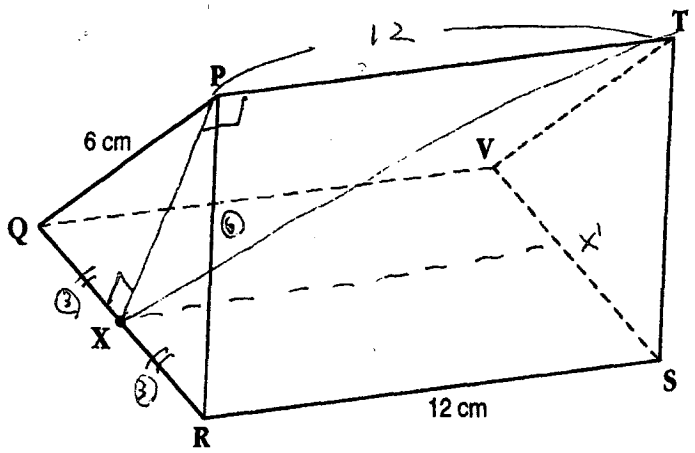
12. a. Given that $A = \begin{pmatrix} 0 & 1 \\ 5 & 6 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 3 \\ -1 & 4 \end{pmatrix}$, find A+2B. **(5 marks)**

$\begin{pmatrix} 0 & 1 \\ 5 & 6 \end{pmatrix} + \begin{pmatrix} 4 & 6 \\ -2 & 8 \end{pmatrix} = \begin{pmatrix} 4 & 7 \\ 3 & 14 \end{pmatrix}$

b. **Figure 4** is a prism with a cross section of an equilateral triangle **PQR**. **PRST**, **PQVT** and **QRSV** are rectangles, such that **RS** = 12 cm, **QP**= 6 cm and **X** is midpoint of **QR.**



$3 \overline{)171}$
$3 \overline{)57}$
19

Figure 4

$PX = 3\sqrt{3}$

$(3\sqrt{3})^2 + 12^2 = TX^2$

$27 + 144 = TX^2$

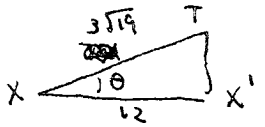
$171 = TX^2$

$TX = 3\sqrt{19}$

Calculate:

(i) **the length of TX** giving your answer correct to 2 decimal places.

(ii) the angle between TX and the base QRSV giving your answer correct to 2 decimal places. **(10 marks)**



$\cos\theta = \frac{12}{3\sqrt{19}} = \frac{4\sqrt{19}}{19}$

$\theta = \boxed{}$

END OF QUESTION PAPER

NB: This paper contains 6 pages.